SLUM AREA SEGMENTATION USING DEEP LEARNING AND TECHNIQUES

Project-I report submitted to
Indian Institute of Technology,
in partial fulfilment of the requirements
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in
Mechanical Engineering

by

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(20ME3AI11)

Under the supervision of

Prof. Adway Mitra

Center of Excellence in Artificial Intelligence

Indian Institute of Technology Kharagpur Spring Semester, 2023-24

Certificate

This is to certify that the project report entitled "**Slum area segmentation using Deep Learning and Techniques**" submitted by **Amit Kumar** (Roll No.: 20ME3AI11) to the Indian Institute of Technology, Kharagpur towards the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (Hons.) in Mechanical Engineering of the Institute, is a record of bona fide work carried out by him under my supervision and guidance during the Autumn Semester of 2023.

Professor Adway Mitra

Date: 27th Nov 2023 Center of Excellence in Artificial

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Kharagpur - 721302, India

Declaration

I certify that:

a) The work contained in this report is original and has been done by me under the guidance of my supervisor.

b) The work has not been submitted to any other Institute for any degree or diploma.

c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

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e) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: 27/11/2023 Amit Kumar

Place: Kharagpur (20ME3AI11)

Acknowledgement

I extend my sincere appreciation to my supervisor, **Prof. Adway Mitra**, and my guide, **Ms**. **Anjali Raj**, for their invaluable support and guidance during my B. Tech thesis project. Their unwavering encouragement, insightful feedback, and constructive criticism have played a pivotal role in shaping my research and assisting me in achieving my academic aspirations. The wealth of knowledge and expertise they bring to the field has served as a constant source of inspiration, fostering a deeper understanding of the subject matter.

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Thank you all for being a crucial part of my academic journey.

Amit Kumar

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Abstract

The process of urbanization has given rise to the proliferation of informal settlements, commonly known as slums, presenting substantial challenges for effective urban planning and policymaking. Leveraging satellite sensing, which offers the capability to capture high-resolution images of urban regions, has emerged as a promising solution for the detection of these slums. In this Bachelor's Thesis Project (BTP), we introduce a deep learning approach specifically tailored for slum detection, utilizing very high-resolution (VHR) optical images obtained from Google Earth. Our methodology involves the manual labeling of images to construct a dataset, subsequently converted into COCO JSON format for training the deep learning model. The proposed approach employs a modified version of the U-Net architecture, HRNET architecture and Attention U-Net, demonstrating its potential to identify slums within specific city sections. This model holds utility for urban planners and policymakers seeking to address slum-related challenges. Future endeavors can focus on enhancing the model's accuracy, extending its applicability to larger urban areas, or adapting it for deployment in various urban environments.

Chapter 1

Slum Segmentation using Remote Sensing: Introduction

1.1 Introduction

Informal settlements represent a widespread global phenomenon, with slums being a prime example of such settlements. Slums, characterized as the most deprived and marginalized form of informal settlements, are inhabited by approximately one-quarter of the world's urban population. The residents of these slums face severe challenges, grappling with the absence of essential resources such as clean water, proper sanitation, electricity, and other critical services.

The genesis of such communities arises from the incapacity of urban management to meet the escalating demand for housing. Consequently, these settlements pose a global concern, given the absence of essential amenities crucial for maintaining a satisfactory quality of life. Over recent decades, international organizations and governments worldwide have embarked on various endeavors aimed at enhancing and revitalizing slum areas. The efficacy of these initiatives hinges significantly on the precision and promptness of information derived from slum mapping and monitoring. Thus, there exists a pressing need for automated, robust, and efficient methods in the realm of slum mapping and monitoring.

This thesis report endeavors to tackle a crucial challenge in attaining the Sustainable Development Goal, specifically the mandate to create cities and human settlements that are inclusive, safe, resilient, and sustainable. The focus of this study is on the pervasive issue of informal settlements, characterized by a lack of legal recognition and insufficient access to fundamental services like water, sanitation, and housing. Common in many developing world cities due to rapid urbanization and a shortage of affordable housing, informal settlements compel people to inhabit makeshift dwellings in unsuitable locations.

The primary objective of this study is to formulate an algorithm utilizing remote sensing data and machine learning techniques for the detection of informal settlements. Leveraging satellite images, the algorithm aims to pinpoint and map areas within a city where informal settlements likely exist. By furnishing precise and current information on the location and extent of these settlements, the algorithm aims to aid government agencies and stakeholders in formulating effective strategies to enhance the well-being of residents in these areas.

Moreover, the effectiveness of slum upliftment and rehabilitation programs hinges on the availability of a spatial database capable of tracking settlement growth rapidly. This is vital for efficient fund allocation and development monitoring. To achieve this, the delineation and mapping of slums, coupled with demographic statistics, can contribute to the creation of a comprehensive database. Such a database can be shared among various development bodies, fostering collaborative efforts. In the realm of machine learning, limited research has been conducted on informal settlements, and the existing studies predominantly utilize Very High Resolution and High-Resolution satellite imagery.

Additionally, the capability to map these settlements equips organizations and NGOs with the means to deliver effective social and economic assistance. This, in consequence, empowers these communities to progress sustainably, affording the inhabitants a significantly improved quality of life. Such progress aligns with multiple Sustainable Development Goals, addressing objectives such as poverty eradication, enhancement of good health and well-being, provision of quality education, access to clean water and sanitation, availability of affordable and clean energy, promotion of sustainable work and economic growth, and facilitation of access to industry, innovation, and infrastructure.

The attainment of these objectives could be facilitated through the utilization of remote sensing imagery and advanced deep learning algorithms to delineate slum boundaries. This investigation specifically assesses the effectiveness of a modified U-net model architecture with various experiments for slum delineation in twelve slum areas situated in the Indian state of Maharashtra: Colaba, Dharavi Bandra, Andheri, Kurla, Ghatkopar, Chembur, Kurla, Malad, Borivali, Bhandup and Mulund. We got the list of the slums present those areas from the Slum rehabilitation authority, Mumbai. Mapping slums poses challenges due to their varied shapes and textures, necessitating the integration of local knowledge and adaptation of slum indicators into maps. Additionally, the absence of ground truth contributes to a shortage of training data, making the mapping process inherently challenging.

a) Satellite Imaginary

b) Masked Image

Figure 1: Satellite imagery and its corresponding masked image for Bandra region

Chapter 2

Data Collection, Data Preparation and Google Earth Pro

2.1 Data Collection

Exploring the effectiveness of the modified U-net model architecture, this study delves into its application for slum delineation in **twelve** slum areas within the Indian state of Maharashtra. The specific targets in Maharashtra include **Colaba, Dharavi Bandra, Andheri, Kurla, Ghatkopar, Chembur, Kurla, Malad, Borivali, Bhandup and Mulund**. We got the list of the slums present in those areas from the **Slum rehabilitation authority**, Mumbai. The challenge in mapping arises from the diverse shapes and textures of slums, necessitating the integration of local knowledge and the adaptation of slum indicators into maps. Additionally, the absence of ground truth compounds the difficulty, resulting in a shortage of training data for accurate mapping.

The data was collected at three different level-wise-

a) Ward

b) Village

c) Taluka

2.1.1 Steps to collect the slum boundaries

1) Visit the Slum reliability authority website (https://sra.gov.in/)

2) Go to Resources and then choose the option for GIS-MIS slum Data under which we have List of slum data 2015.

3) Here, we have a list of slum areas representing the twelve talukas of Mumbai city.

4) After that, we have searched down all the entries on "**Google Earth Pro software**" for high level resolution (1920px by 1080px) original satellite imaginary at the eye altitude in the range of 980 ft to 1400 ft.

5) We have used "**MakeSense.AI**" software for the manual annotation of slum areas presented in the satellite imaginary and import images as COCO JSON format.

6) Final segregation of slum areas using masks from the JSON file using COCO API and another way by using OpenCV.

7) The code is available at:

2.2 Data Preparation

For the individual latitude-longitude created for slum files, we marked slum mask image as 1 or 0 corresponding to each pixel:

a) 1 indicates slum.

b) 0 indicates non-slum

2.2.1 Converting the JSON format file into masked Image

2.2.1.1 JSON format file

JSON, or JavaScript Object Notation, is a lightweight data interchange format that is easy for humans to read and write, and easy for machines to parse and generate. It is often used to transmit data between a server and a web application, as an alternative to XML.

In the context of annotation in computer vision and machine learning, a JSON file generated by platforms like makesense.ai contains information about the annotations made on an image. This information typically includes details about description, images information (id, width, height, and file name), annotation (id, image_id, category_id, segmentation, bounding box, area) and categories of objects or regions of interest within the image. Here, segmentation stores the coordinates of the polygon which contains the slum area in any image. Each JSON format file contains the information of 50 images.

For example, if you annotate objects in an image to train a machine learning model, the JSON file might include information about the bounding boxes, polygons, or other shapes drawn around each annotated object. It can also include class labels, attributes, or any additional information related to the annotated regions.

COCO API can handle images with multiple classes, which is common in real world scenarios to prepare masked images from the JSON file.

Many popular deep learning frameworks and libraries provide built-in support for COCO format, making it easy to load and process the data for training and evaluation. The COCO dataset is one of the largest and most well-known datasets in the computer vision community.

2.2.2 Images in Google Earth Pro

Google Earth Pro, is an advanced version of the popular Google Earth application, designed to meet the needs of professionals and users requiring additional features. One notable capability of Google Earth Pro is its advanced measurement tools, allowing users to measure distances, areas, and volumes directly on the map, catering to fields like construction, surveying, and mapping. Another distinctive feature is its high-resolution printing function, enabling users to create detailed images and print them at resolutions of up to 4,800 pixels, ideal for presentations and reports.

Moreover, Google Earth Pro provides enhanced map-making tools, offering users the ability to customize maps with legends, titles, and other elements. The application supports the import of GIS data, facilitating the overlay of custom datasets onto the 3D globe for analysis. Users can create dynamic presentations or virtual tours using Google Earth Pro's movie-making capabilities, recording their navigation within the application. Additionally, access to historical imagery allows users can explore changes in a location over time. The application offers higher resolution images and often boasts faster loading times compared to the standard version.

Fig: Image taken using Google Earth Pro

2.2.3 Annotation and MakeSense.AI

MakesSense. AI is a widely used software or platform. It is used to create mask which is used for both object detection and image recognition. It provides various options for segment out the area/ region of interest from the original image. We can create many areas of interest as per our requirement. We can then export the information of the annotated image in the form many formats and some of the Well known formats about which we are aware about are COCO JSON format, VGG JSON, CSV format. We can annotate many images at once and store the data of those images in a single JSON file in a very ordered way.

Fig: Annotation using MakeSense.AI

Chapter 3

Models and Results

3.1 U-Net Model

3.1.1 Introduction

The U-Net, conceived in 2015 by Olaf Ronneberger, Philipp Fischer, and Thomas Brox at the University of Freiburg, Germany, stands as a pivotal deep-learning model architecture originally crafted for biomedical image segmentation. This architecture, which has gained widespread recognition, takes its name from the distinctive shape of its structure, resembling a "U."

At its core, the U-Net model features an encoder-decoder structure, making use of convolutional neural network (CNN) principles. This architecture is particularly tailored to address image segmentation challenges, wherein the goal is to partition an image into various segments and precisely identify the specific objects or features within each segment. The U-Net's versatility and efficacy have led to its application in diverse fields beyond biomedical imaging, showcasing its adaptability in tackling complex segmentation tasks across various domains.

3.1.2 Model Architecture

The architecture's unique structure, resembling a "U," consists of an encoder-decoder framework, each with specific functionalities.

1. *Encoder Structure*: The encoder section of the U-Net is responsible for capturing contextual information from the input image. It comprises a series of convolutional and pooling layers that progressively reduce spatial dimensions while extracting high-level features. This down sampling process helps the model grasp broader patterns and context within the image.

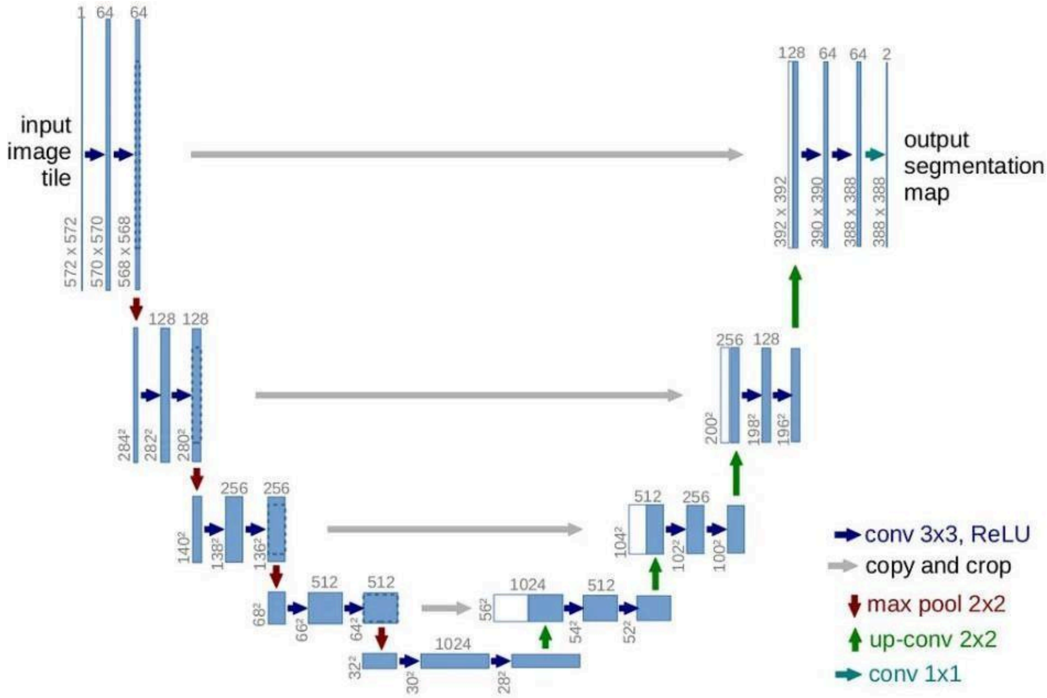
2. *Bottleneck*: Positioned between the encoder and decoder, the bottleneck or central layer serves as a bridge between the spatial information captured by the encoder and the detailed segmentation performed by the decoder. It acts as a bottleneck by compressing the feature representation.

3. *Decoder Structure*: The decoder section, mirroring the encoder's structure, is responsible for up sampling the feature representation back to the original image dimensions. This involves a series of up sampling and convolutional layers to recover spatial details lost during the encoding process. Skip connections, which directly link corresponding layers of the encoder and decoder, facilitate the transfer of detailed information across different scales.

4. *Skip Connections*: A key innovation in the U-Net architecture is the use of skip connections. These connections link layers from the encoder to the corresponding layers in the decoder. This enables the model to retain detailed information during the down sampling and up sampling processes, aiding in the precise localization of features.

5. *Final Layer*: The final layer of the U-Net is a 1x1 convolutional layer followed by a SoftMax activation function. This layer produces the segmentation map, where each pixel is assigned a class probability, determining the presence of specific objects or features in the corresponding region.

The U-Net's architecture has proven highly effective in image segmentation tasks, not only in the biomedical domain but also in various other fields where accurate delineation of objects within images is crucial. Its adaptability and success have made it a foundational model in the realm of convolutional neural networks.



UNet architecture

In summary, U-Net is a popular deep learning model architecture designed for image segmentation tasks. Its encoder-decoder structure with skip connections allows for accurate segmentation results by preserving spatial information and extracting relevant features from the input image.

3.1.3 Model Training, Testing and Parameters

1. **Parameters for the model.**: Within the parameters of this function, key specifications encompass the determination of the input image tiles' band count, the definition of the **number of classes**, and the establishment of the **number of epochs**. The subsequent code section encapsulates the definition of a U-Net deep learning model. Notably, the chosen model architecture is a customized iteration of the U-Net, adapted from Wang et al.'s work in 2019.

2. The model initiation involves the coding of the **encoder** component, which adopts the CNN model as its foundation. Subsequently, dedicated sections of code for the U-Net model follow suit. The U-Net model itself commences by specifying the dimensions of input images, succeeded by the incorporation of a convolutional layer featuring 16 filters, each with a size of 3x3.

3. Sequentially, a **batch normalization layer** is applied, succeeded by a **rectified linear unit** (**ReLU**) activation layer. This is followed by an additional convolutional layer featuring 16 filters, each sized at 3x3. The layer sequence continues with a batch normalization layer, a ReLU activation layer, and culminates with a max-pooling layer of dimensions 2x2. It's noteworthy that the output from the preceding layer serves as the input for the subsequent layer, fostering information flow continuity.

4. Similar codes sections repeat 4 times with exponential growth of convolutional layers from **16 to 64, 128, 256 and 512**. In this way the encoder part of the U-Net model completes.

5. The fifth section of codes is also similar to the other with convolutional layers with 512 filters but does not have a max pooling layer. This part is also known as “**bottleneck**” from where the decoder part of the U-Net model starts.

6. Subsequently, the code encompasses sections dedicated to constructing the second component of the model known as the **decoder**. In the decoder section, code blocks are dedicated to **transposing convolutional layers**. This implies that these sections effectively replicate the encoder's architecture but in the reverse direction, employing transposition as a transformative operation to achieve this inverse structure.

7. **Batch normalization** helps the model to regularize and learn faster. During up sampling the feature space from the encoder part also known as skip connections join with the decoder part to learn the intricate properties of the image.

8. The function definition for the model is succeeded by a segment of code dedicated to functions for metrics, providing insights into the model's behavior. Following this metrics section, there is another code segment that outlines the functions responsible for optimization. This sequential arrangement highlights the comprehensive structure of the code, where model definition, metric assessment, and optimization strategies are systematically organized.

9. The line that compiles the line of codes features **a loss function** which is **a focal loss, this** is due to the problem we want to solve by classifying our image in two classes (informal settlements and non-informal settlements). If the user prefers to implement multi-classes classification, the loss function should be replaced by the categorical cross-entropy.

10. **Model Evaluation:** Then comes evaluating the model performance. Those blocks are composed of the two code sections for plotting training curves (accuracy and loss) and the section of codes for evaluating the model performed on the validation dataset.

11. **Predictions:** The last part is made of a code section for inferencing the model on the image tiles for prediction.

3.2 Res-Net Model

ResNet50, short for Residual Network with 50 layers, is a deep convolutional neural network architecture introduced by Microsoft Research in 2015. ResNet50 is part of the ResNet family, which aimed to address the challenge of training very deep neural networks by introducing residual learning blocks. The fundamental innovation in ResNet is the use of residual or skip connections that bypass one or more layers. This facilitates the creation of deeper networks by mitigating the vanishing gradient problem. In the ResNet50 architecture, the network consists of 50 layers, including residual blocks with skip connections. It is characterized by a stack of residual blocks, each containing multiple convolutional layers and shortcut connections. The architecture employs global average pooling and a fully connected layer for the final classification. ResNet50 has proven highly effective in various computer vision tasks, particularly in image classification, due to its ability to efficiently train very deep networks while maintaining performance gains.

Integrating ResNet50 as the encoder in a U-Net model represents a strategic augmentation of the feature extraction process. In this configuration, the ResNet50, renowned for its deep architecture and residual connections, supplants traditional convolutional layers in the U-Net's encoder section. Leveraging a pre-trained ResNet50 model, the initial layers are typically frozen to retain their learned features. The output from these layers is then seamlessly integrated into the U-Net architecture, enhancing its capacity to capture intricate hierarchical features from input images. This fusion of ResNet50 and U-Net is particularly advantageous in image segmentation. tasks, as ResNet50's prowess in learning high-level features contributes to improved semantic understanding, ultimately enhancing the model's precision in delineating objects or regions of interest. Fine-tuning the integrated model, if necessary, ensures adaptability to specific datasets and tasks.

The ResNet50 is used as the encoder of the UNET which is pretrained on the Image-Net Dataset. Further, we have used freeze the weight of the encoder part and then we train our model on us Dataset to find all the weights and biases for the decoder part. ImageNet is a large-scale dataset widely recognized in the computer vision community, primarily used for image classification tasks. However, when it comes to image segmentation, ImageNet itself may not be the most suitable choice due to its annotations focusing on object-level labels rather than pixel-level segmentation masks.

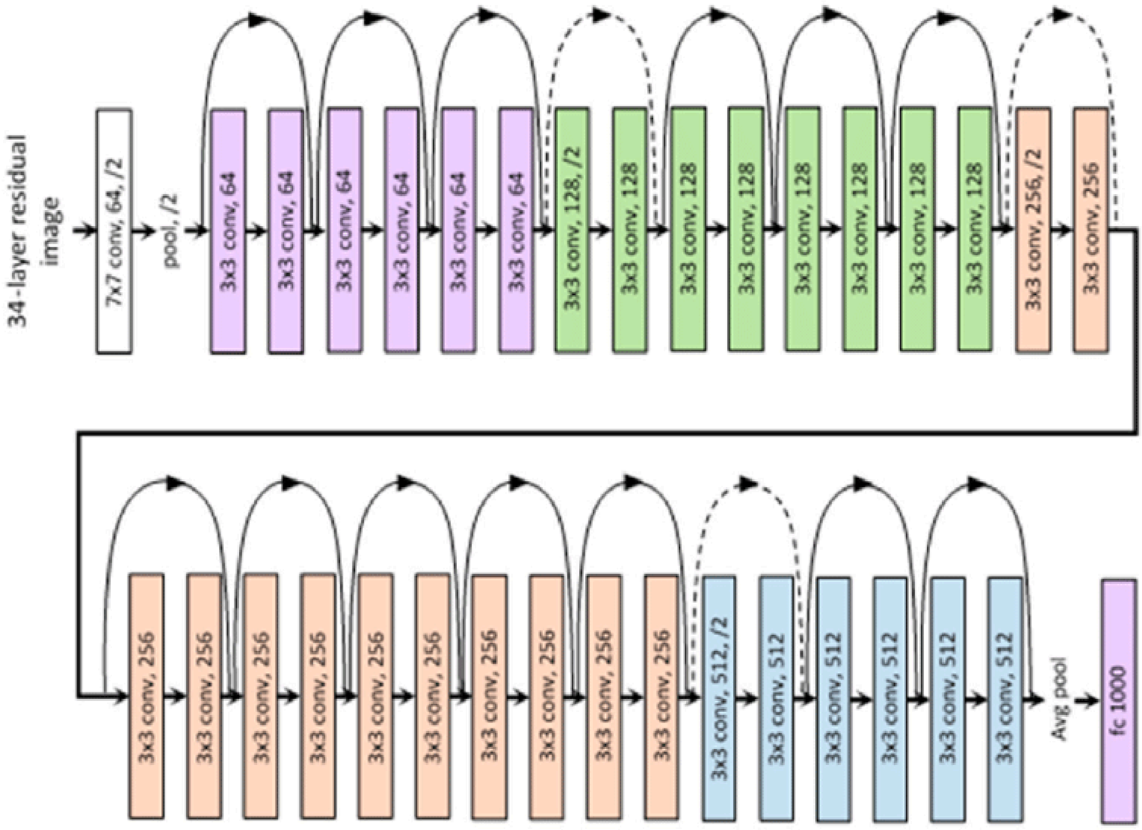


Fig: ResNet50 Architecture

We have used UNet with ResNet50 as encoder pretrained on Image net dataset model to find out the best optimizer among (**Adam, Adadelta, Adagrad, RMSPROP, SGD**) and loss functions (**Focal Loss, Hinge Loss, Binary-cross Entropy, MSE Loss**).

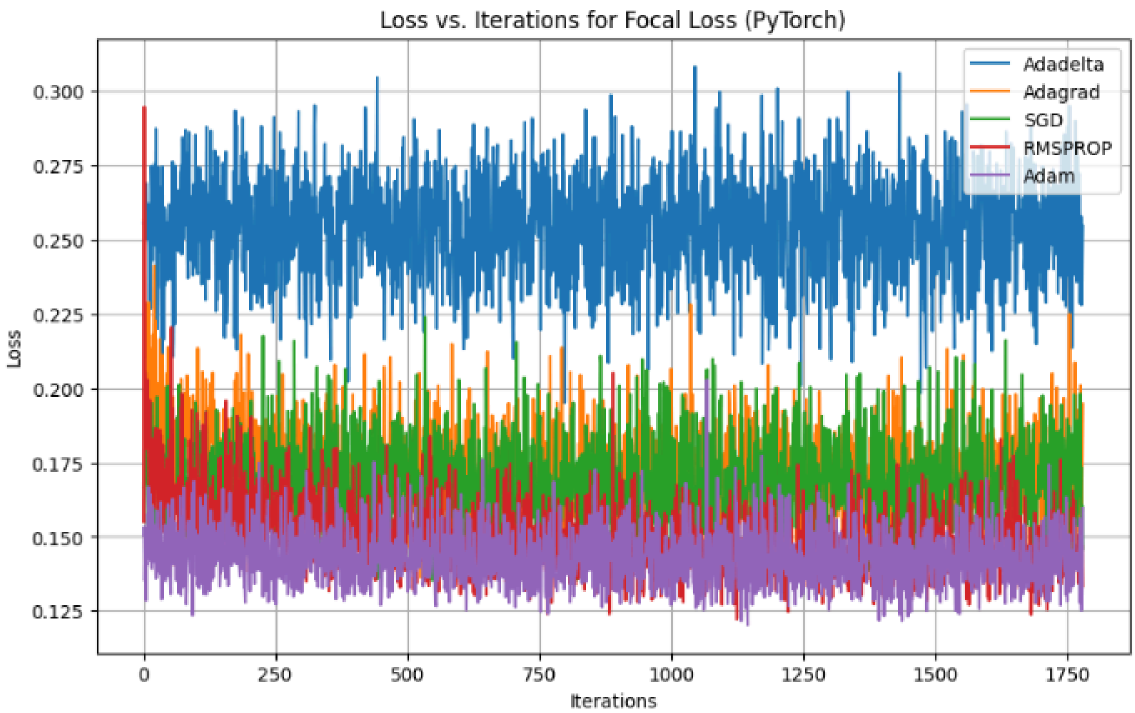
3.2.2 Model parameters and Graphs:

When trained on **Forest Dataset** and checked for other hyperparameters.

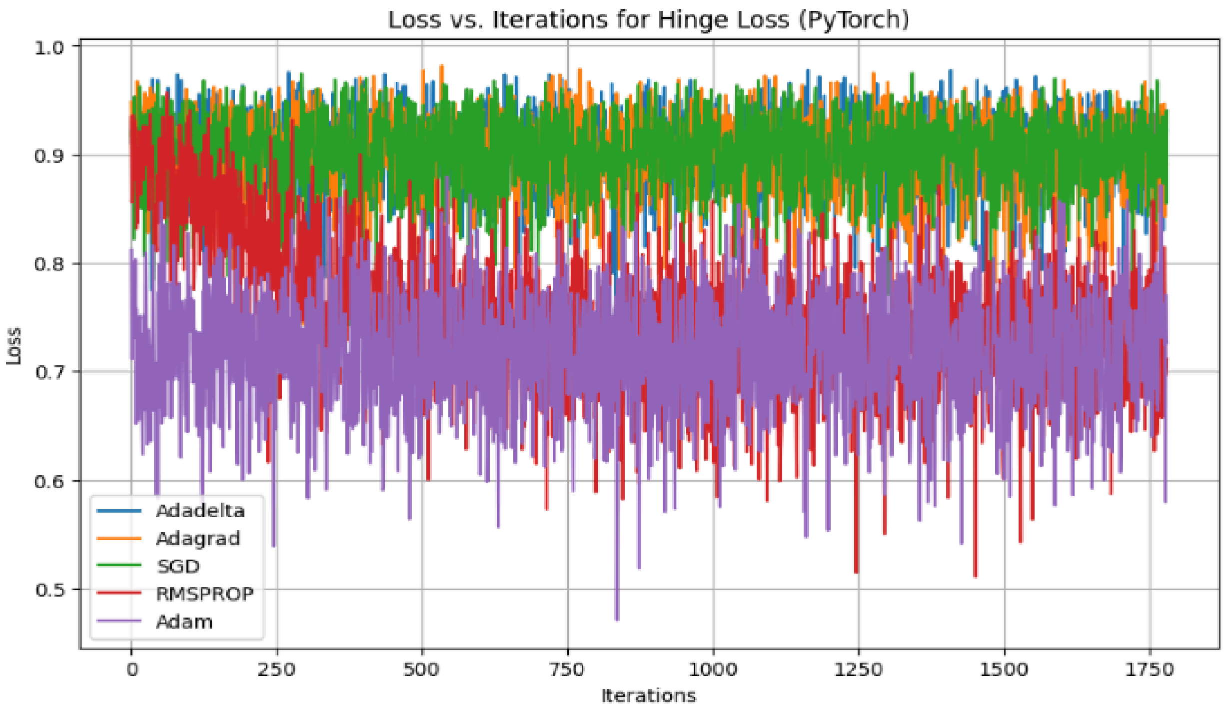
- Number of_Epochs:10
- Batch size: 16
- Learning Rate: 1e-4
- Image Height: 256
- Image Width: 256
- Train Size: 2838
- Validation Size: 710

Graphs:

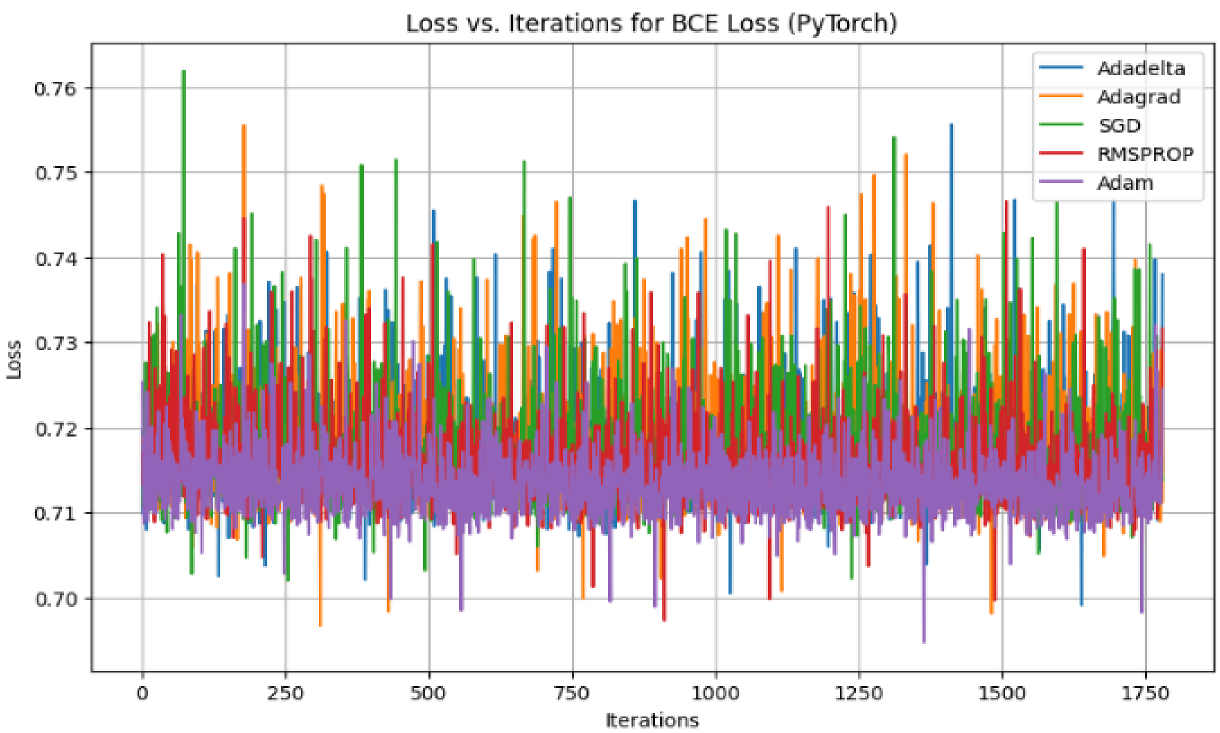
Focal loss vs iterations for all the optimizer:



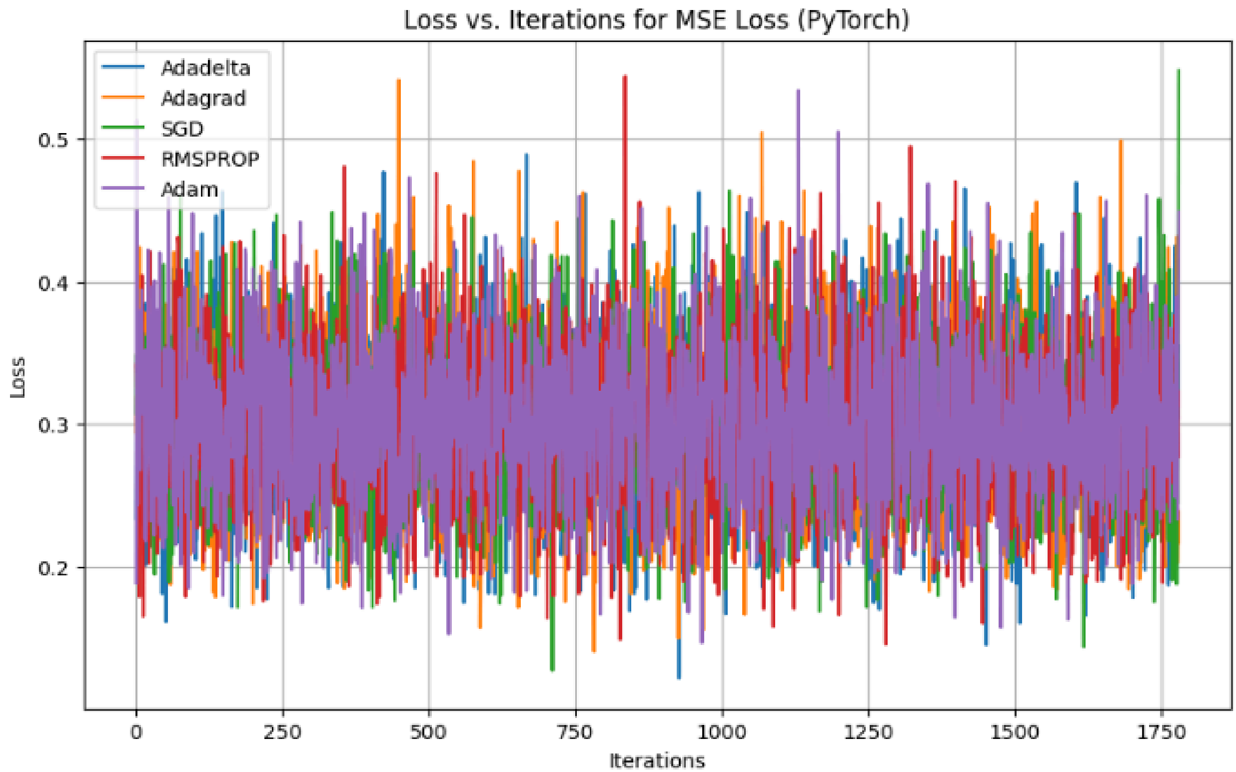
Hinge Loss vs Iterations for all optimizers:



BCE Loss vs Iteration for all optimizer:



MSE Loss vs Iterations for all optimizers:



Conclusion: From the above plots, we have observed that we got best metrics corresponding to **Focal Loss** and **Adam Optimizer**. Hence, we have used Focal loss and Adam optimizer for training our further models.

3.3 UNET Model on Forest Dataset implemented from scratch

Description of forest dataset: Automatic categorization and segmentation of forest cover is of great importance for sustainable development and urban planning. The importance of forests cannot be under-estimated as they support the flow of essential ecosystem services such as fibre, energy, recreation, biodiversity, carbon storage and flux and water. Forests are also important areas to identify before commencing any kind of industrial activity that requires on field work. Satellite or remote sensed images can be used to identify and segment the forest cover regions in the image and get a clear idea about how much land does forest cover. This problem is defined as a binary segmentation task to detect forest areas.

This dataset contains 5108 aerial images of dimensions 256x256.

Training and Testing Split: 0.1

Train Size: 4597

Val Size: 511

We have just run our model on this dataset which is publicly available on Kaggle, just to check whether the implemented model is working fine or not.

3.3.1 Model Parameters and Plots

When trained on **Forest Dataset** and checked for other hyperparameters.

- NUMBER_EPOCHS = 12
- Batch size: 32
- Learning Rate: 1e-4
- Image Height: 256
- Image Width: 256

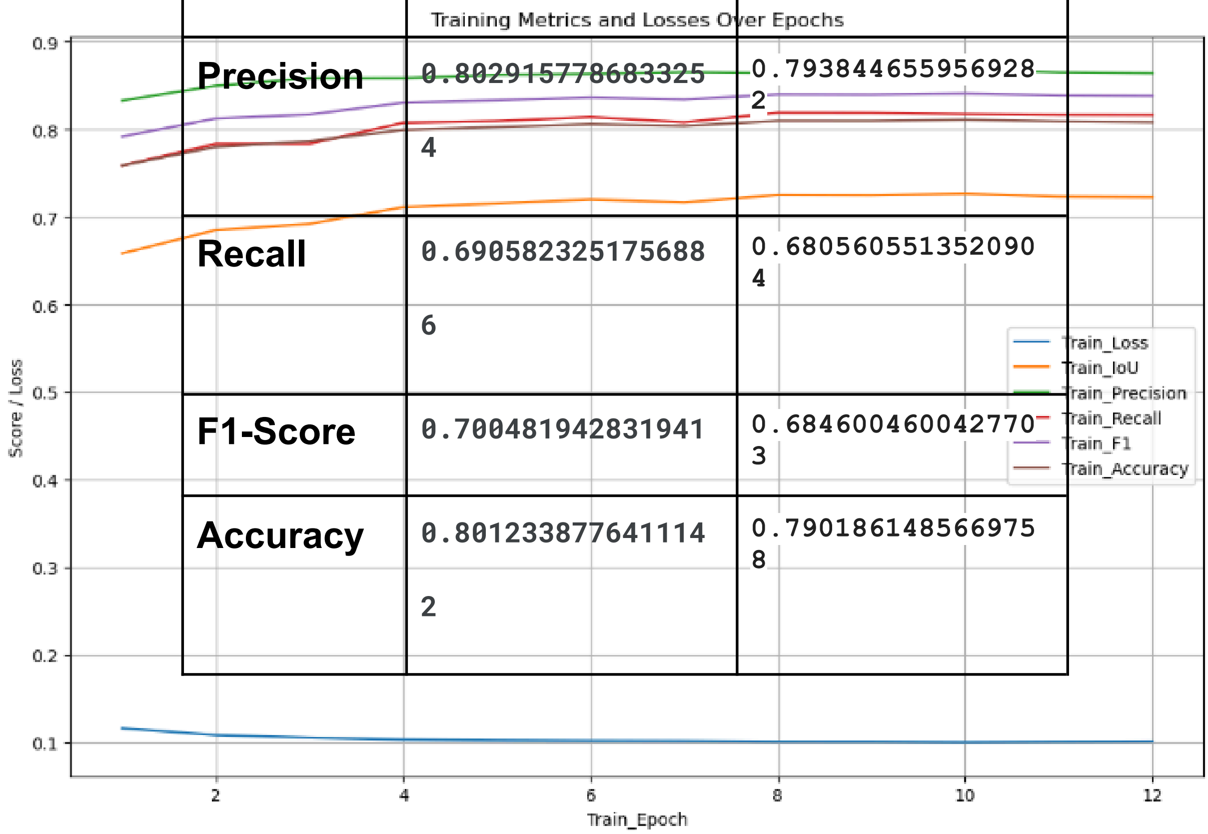
Training:

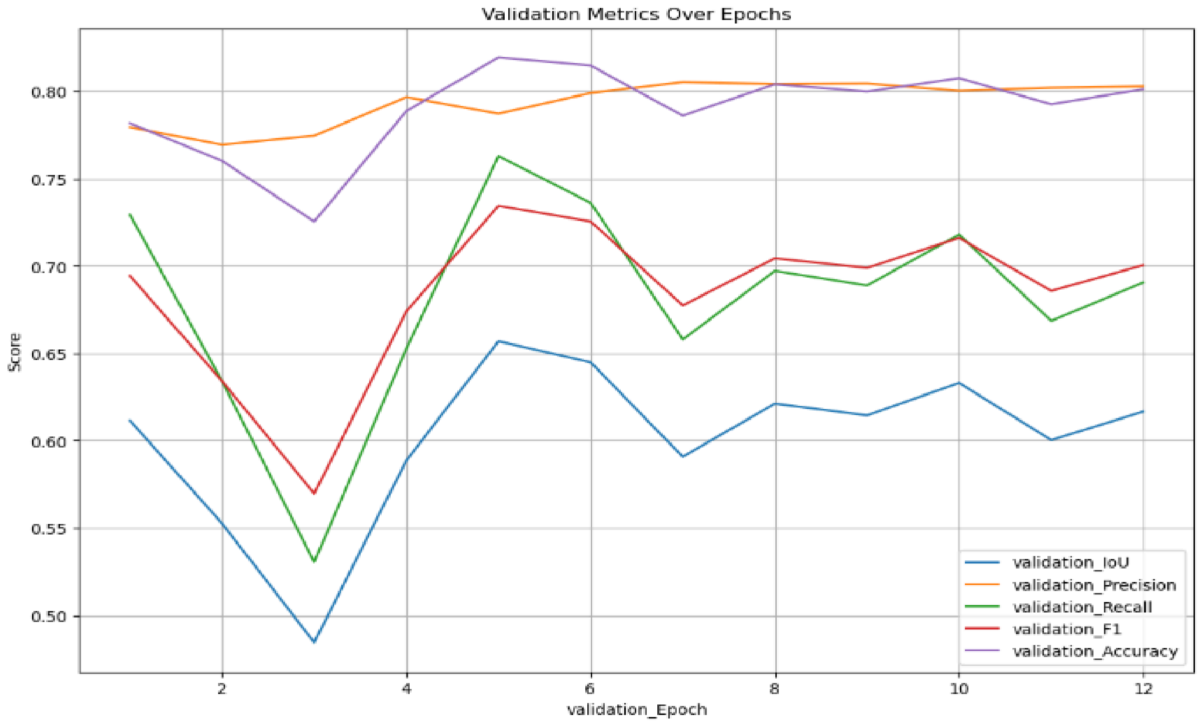
Training	Final Val	Average Val
Loss	0.10135573143553403	0.10377412667946406
IOU	0.7227035253473302	0.7101781204775013
Precision	0.8643540177562743	0.8598833111307101
Recall	0.816464339267975	0.804535935474347
Accuracy	0.808211	0.801234
F1-Score	0.8383003025058201	0.8294679393627248

Validation:

Validation	Final Val	Average Val
IOU	0.6166124624508018	0.6012564980113367
Precision	0.8029157786833254	0.7938446559569282
Recall	0.6905823251756886	0.6805605513520904
F1-Score	0.700481942831941	0.6846004600427703
Accuracy	0.8012338776411142	0.7901861485669758

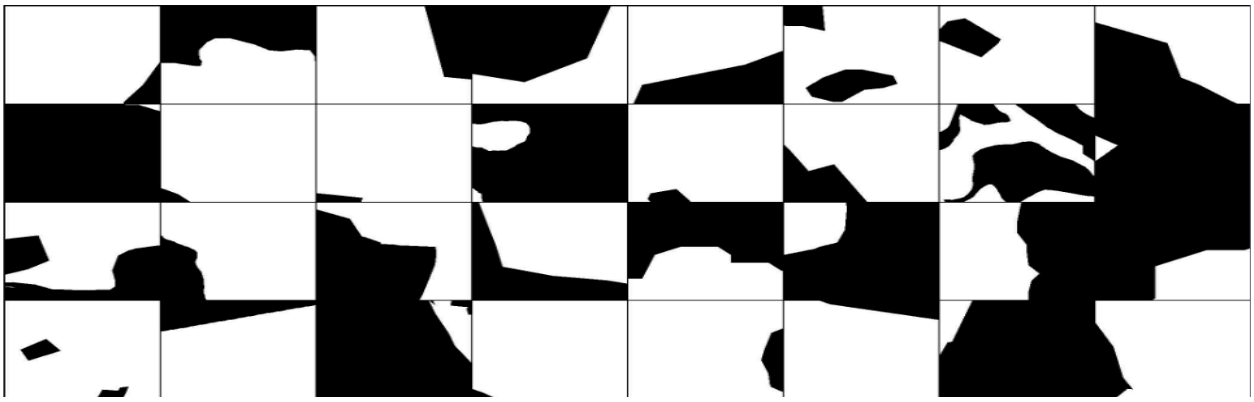
Graphs:



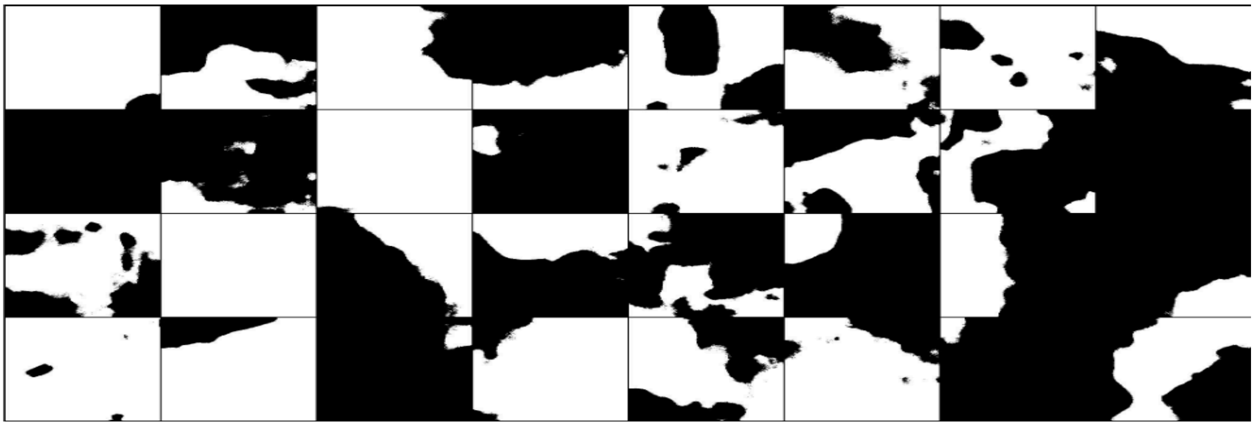


3.3.2 Original and Predicted Masks

Original Mask:



Predicted Mask:



3.4 UNET Model on Slum Dataset implemented from scratch

3.4.1 Model parameters and Metrics Table:

When trained on **Slum Dataset** and checked for other hyperparameters.

- Number of_Epochs:15
- Batch size: 16
- Learning Rate: 1e-4
- Image Height: 256
- Image Width: 256
- Train Size: 2838
- Validation Size: 710

Training:

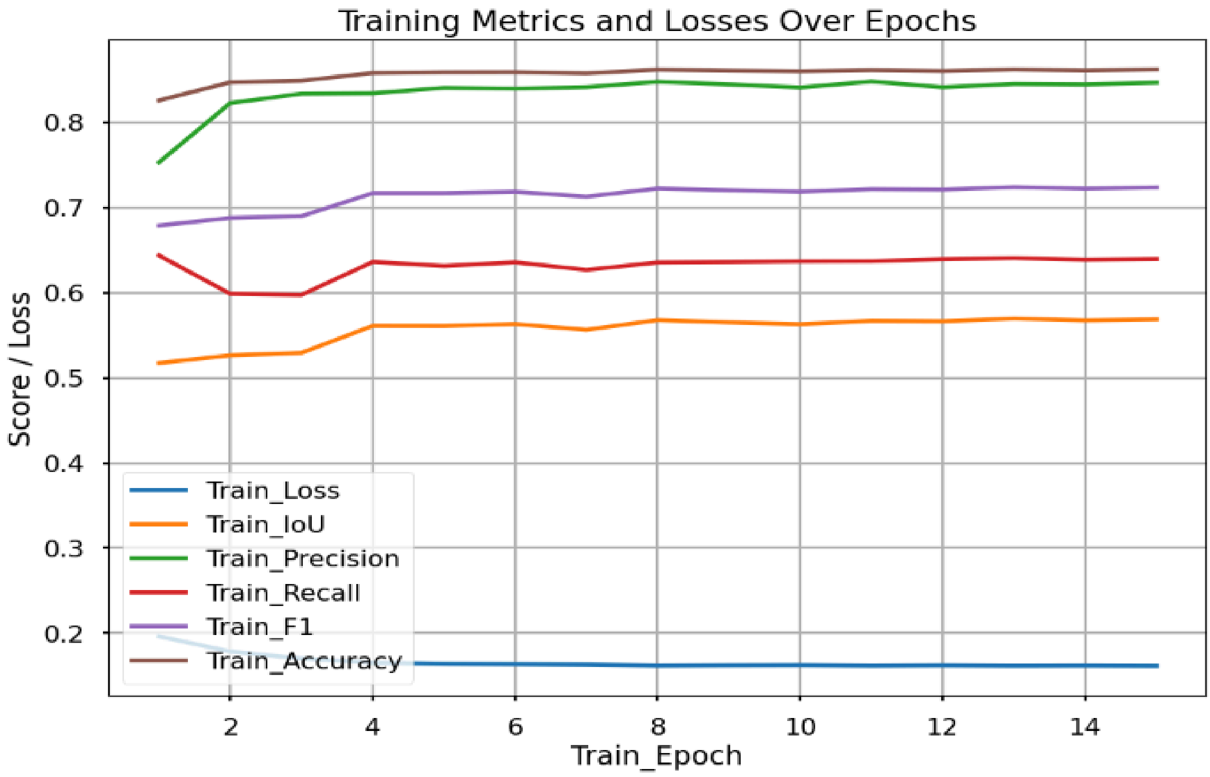
Training	Final Val	Average Val
Loss	0.16171999761227812	0.16654365693362022
IOU	0.5686634915620401	0.5566380467757983

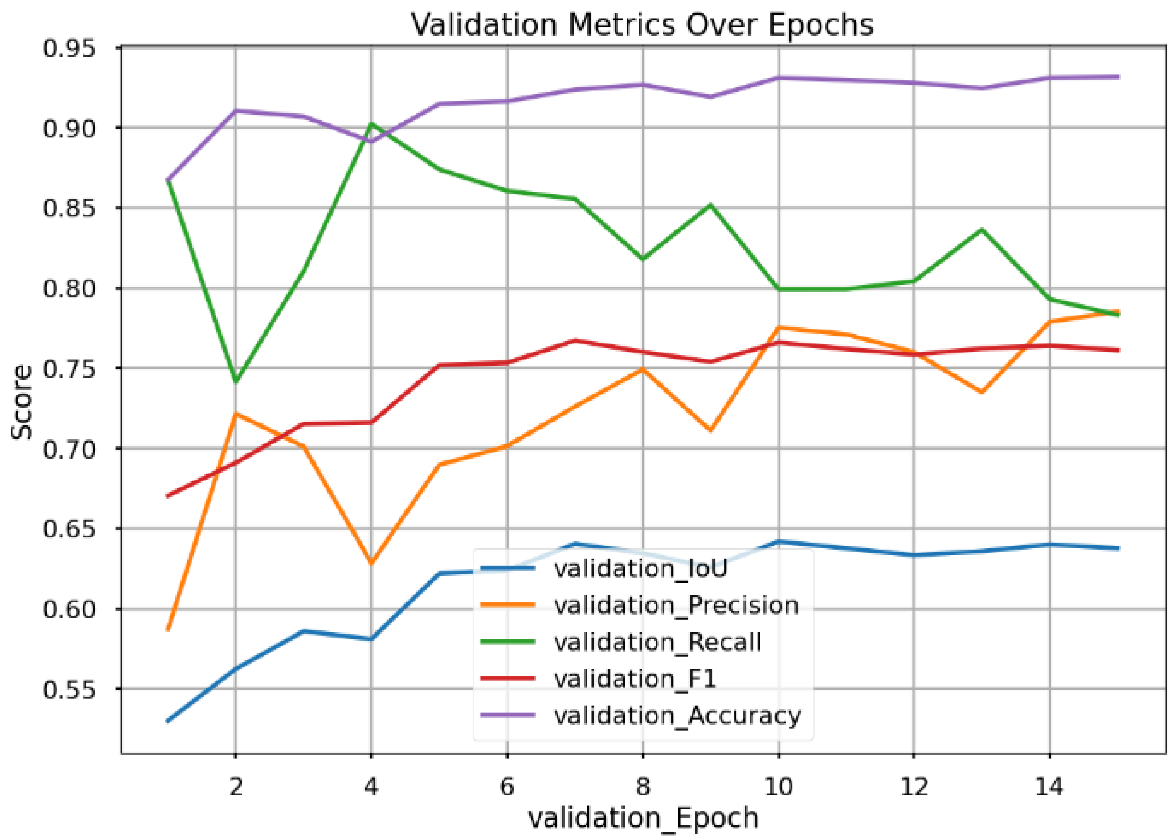
Precision	0.8465993274450812	0.8349072950992006
Recall	0.6395850165453681	0.6315628477415703
F1 - score	0.7235773585304585	0.7130114550255087
Accuracy	0.862241	0.849231

Validation:

Validation	Final Val	Average Val
F1 - Score	0.761317065376618 9	0.7436042919449413
IOU	0.637830176234715 4	0.6156258409476802
Precision	0.785271059163082 3	0.7214403376009578
Recall	0.783170702113513 1	0.8264534728985368
Accuracy	0.931719755790603	0.9168841466769367

3.4.2 Graphs and Masks:





Original and Predicted Masks for UNET from scratch:

Original Mask:

Predicted Mask:

3.5 UNET model with ResNet50 as the encoder pretrained on the ImageNet dataset:

3.5.1 Model parameters and Metrics Table:

When trained on **Slum Dataset** and checked for other hyperparameters.

- Number of_Epochs:15
- Batch size: 16
- Learning Rate: 1e-4
- Image Height: 256
- Image Width: 256
- Train Size: 2838
- Validation Size: 710

Training:

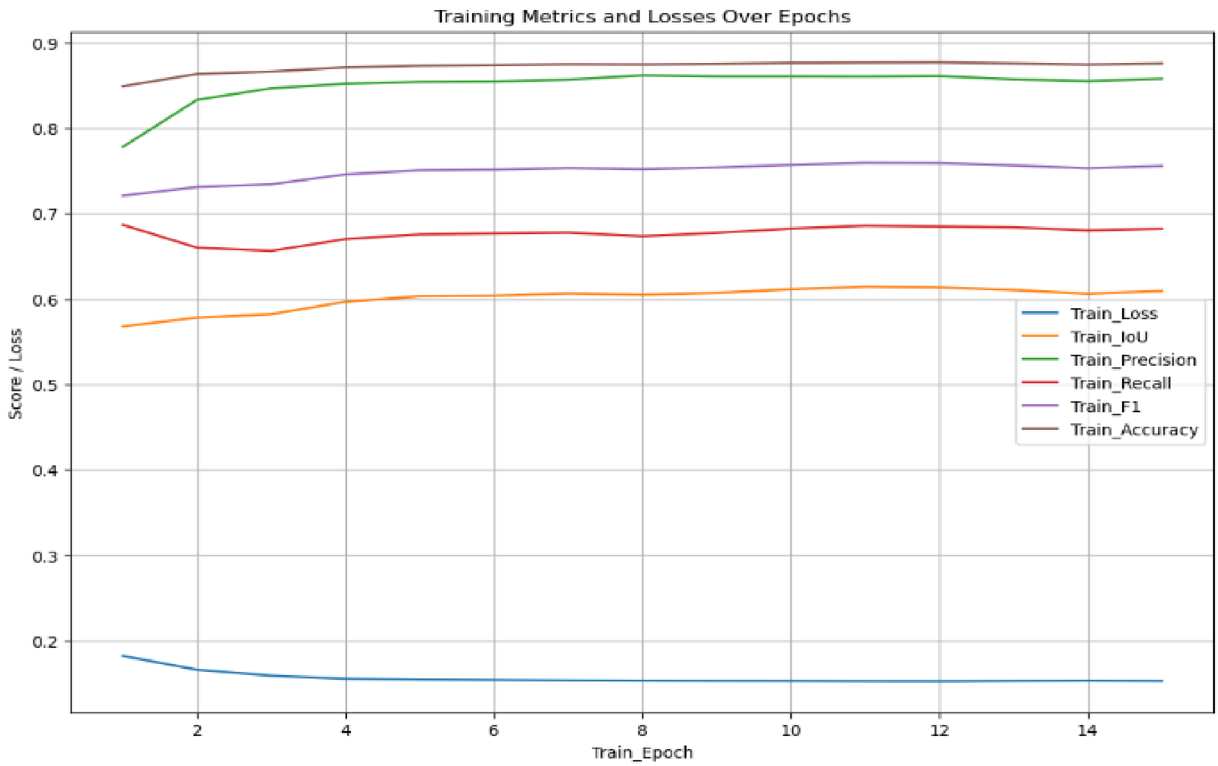
Training	Final Val	Average Val
Loss	0.15264741737353668	0.1563533278049601
IOU	0.60912188871678	0.600839187892621
Precision	0.857534713543067	0.8497133302562274
Recall	0.6820642527467212	0.6767127175203704
F1_Score	0.7555368237317339	0.7486948195401464
Accuracy	0.875480	0.850345

Validation:

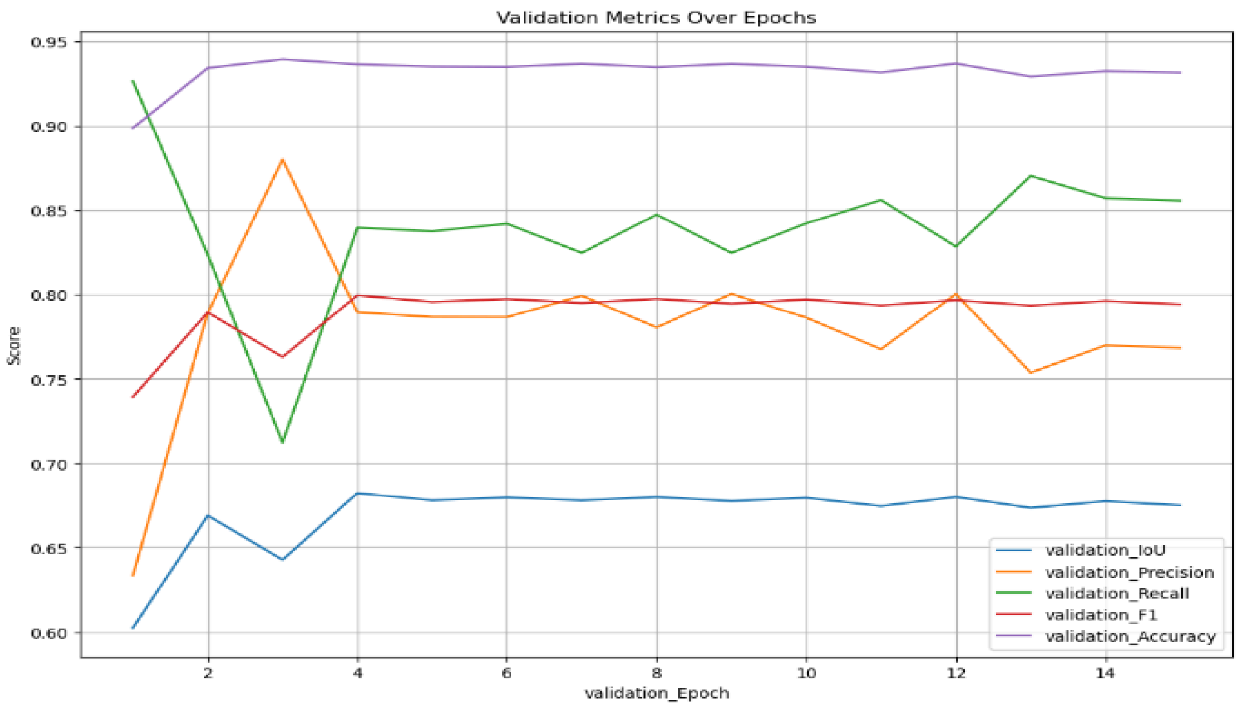
Validation	Final Val	Average Val
F1_Score	0.7941789763398227	0.7895380828432247
IOU	0.6750000610948864	0.6699686790982348
Precision	0.7684059812647547	0.779473048054427
Recall	0.8556791261945653	0.8392126686966378
Accuracy	0.931579374931228	0.9322925148547535

3.5.2 Graphs and Masks:

Training:



Validation:



Original and Predicted Mask for ResNet50 as encoder:

Original Mask:

Predicted Mask:

Chapter 4

HRNet Model and Results

Introduction:

HRNet, short for High-Resolution Network, is a state-of-the-art deep learning architecture designed for high-resolution visual recognition tasks. Introduced by researchers at the Beijing Lab of Megvii Technology in 2019, HRNet stands out for its ability to maintain high-resolution representations throughout the network, avoiding the down sampling common in many traditional architectures. This preservation of fine-grained details at multiple scales allows HRNet to excel in tasks such as human pose estimation, semantic segmentation, and object detection, where retaining high-resolution information is crucial. HRNet's innovative design involves parallel subnetworks that process information at different resolutions and iteratively refine representations, enabling the model to capture both global context and fine-grained details simultaneously. Its effectiveness in handling high-Resolution visual data has made HRNet a popular choice for various computer vision applications, particularly in scenarios where precise localization and detailed information are essential.

Model Architecture:

HRNet, or High-Resolution Network, is a deep neural network architecture designed for high-resolution visual tasks, particularly in the field of human pose estimation. The key characteristic of HRNet is its ability to maintain high-resolution representations throughout the network, enabling it to capture fine-grained details in images.

Here's a simplified explanation of the HRNet architecture:

1. **High-Resolution Representation Learning:**

HRNet starts by processing the input image at its original resolution. Many traditional networks down sample the input resolution early in the network to reduce computational complexity. However, HRNet takes a different approach by maintaining high-resolution representations from the beginning.

2. Multi-Scale Feature Fusion:

HRNet employs a multi-scale feature fusion strategy to combine information from different resolutions effectively. This is crucial for capturing both global context and fine details in the image. The network has parallel branches that process the features at different resolutions, and These branches are interconnected to share information.

3. High-Resolution Skip Connections:

Skip connections are used to connect features at different scales directly. This helps in preserving spatial information and allows the network to recover details lost during down-sampling.

4. Hourglass Structure:

The HRNet architecture often adopts an hourglass structure, which is a recurrent design that repeats encoding and decoding stages. This structure helps in refining the features and capturing hierarchical information at different scales.

5. Intermediate Supervision:

HRNet may include intermediate supervision, where predictions are made at multiple intermediate stages of the network. This helps in training the network more effectively by providing supervision at different levels of abstraction.

6. Task-Specific Heads:

The final part of the network typically includes task-specific heads that generate predictions based on the high-resolution features learned by the network. In the context of human pose estimation, these heads would output the coordinates of key points on the human body.

7. Training Strategy:

HRNet is often trained end-to-end using annotated datasets for the specific task it is designed for. The loss function used during training is task-dependent; for example, mean squared error may be used for regression tasks like human pose estimation.

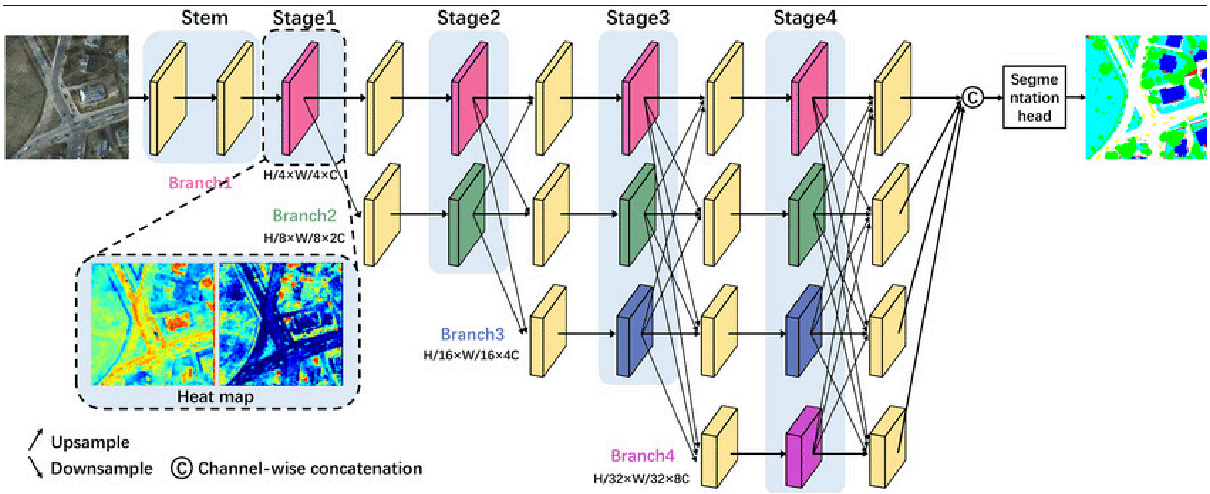


Fig: HRNET Architecture

HRNet (High-Resolution Network) is a deep neural network architecture designed for high-resolution computer vision tasks, particularly semantic segmentation. The number "48" in "HRNet 48" refers to the number of high-resolution feature maps or the width of the network.

HRNet is characterized by its ability to maintain high-resolution representations throughout the entire network, as opposed to down-sampling the resolution early in the architecture. This approach helps in capturing fine-grained details and maintaining spatial information, making it is well-suited for tasks where precise localization is crucial.

Model Training, Validation and Hyperparameters:

When trained on **Slum Dataset** and checked for other hyperparameters.

- Number of_Epochs:15
- Batch size: 16
- Learning Rate: 1e-4
- Image Height: 256
- Image Width: 256
- Train Size: 2838

- Validation Size: 710

Metrics Table, Graphs and Masks:

Tables:

Training:

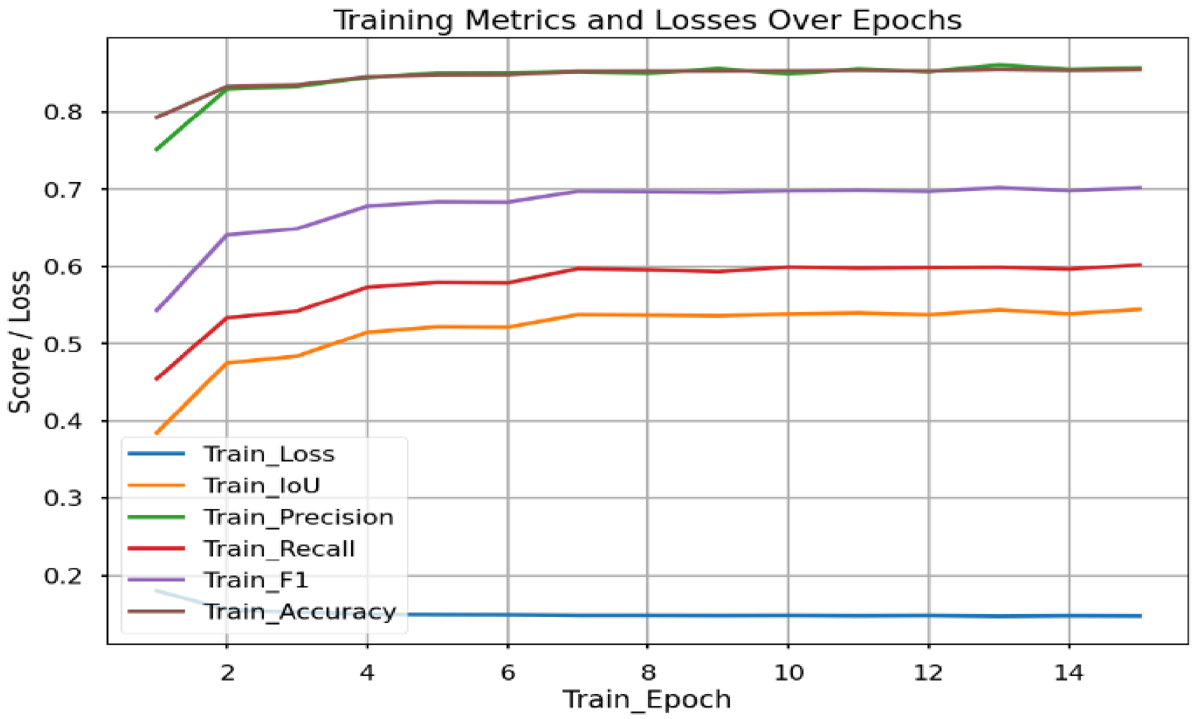
Training	Final Val	Average Val
Loss	0.14743848086408015	0.15101857808198818
IOU	0.5443597481593979	0.5170111262243613
Precision	0.8570809273079699	0.8435118345184424
Recall	0.6020805034925913	0.57602366832559
F1_Score	0.7022975116265356	0.6776749002501288
Accuracy	0.855422	0.843321

Validation:

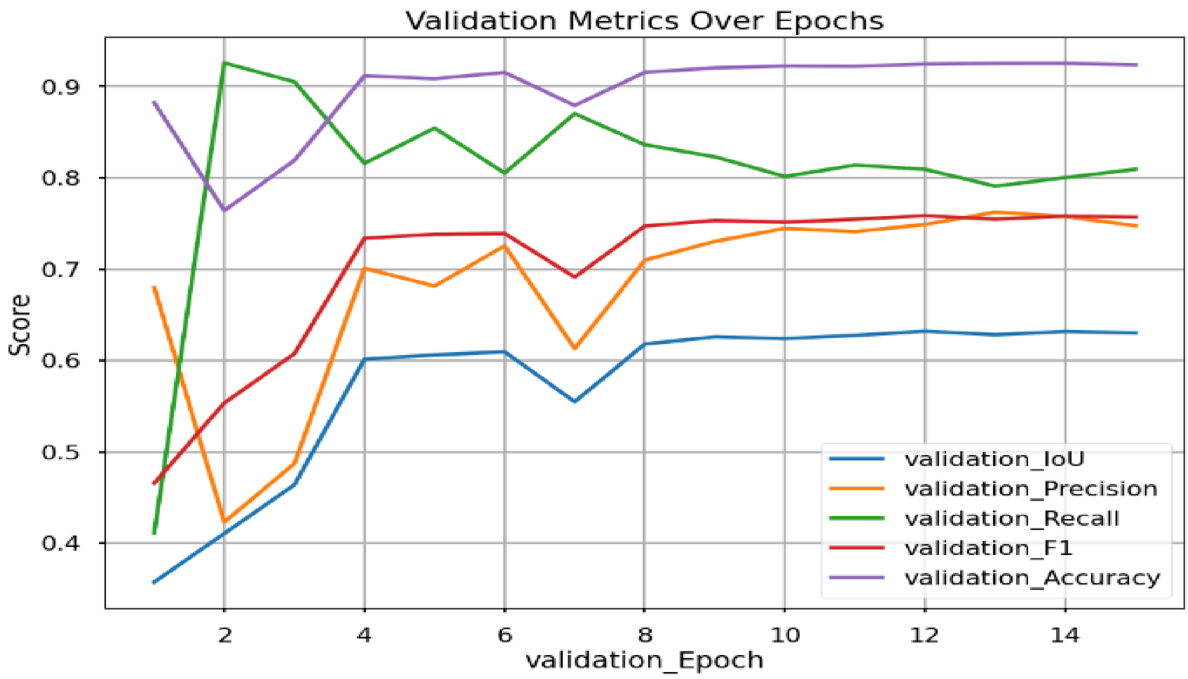
Validation	Final Val	Average Val
F1_Score	0.7570561156214068	0.7042782562360378
IOU	0.6303501523527772	0.574820522667881
Precision	0.7476004924292327	0.6836438874016512
Recall	0.8094193876282992	0.8048691742501923
Accuracy	0.9237862385494608	0.8973762404750771

Plots:

Training:



Validation:



Original vs Predicted Mask:

Original Mask:

Predicted Mask:

Chapter 5

Attention UNet Model and Results

Introduction:

The Attention U-Net is an advanced variant of the U-Net architecture, integrating attention mechanisms for improved feature weighting and contextual understanding. By strategically placing attention blocks within the network, it selectively emphasizes or suppresses regions of feature maps, enhancing its ability to capture long-range dependencies. This variant is particularly effective in image segmentation tasks where precise delineation of object boundaries and intricate details are crucial. The Attention U-Net retains the basic U-Net structure, utilizing an encoder-decoder design with skip connections. Its applications span medical image analysis, such as semantic segmentation in medical imaging, and other computer vision domains where nuanced contextual information is essential for accurate segmentation.

Model Architecture:

The Attention U-Net architecture extends the classic U-Net design by incorporating attention mechanisms, enhancing its ability to capture contextual information and improve feature weighting for image segmentation tasks. Here is a detailed breakdown of the Attention U-Net architecture:

1. Encoder-Decoder Structure:

Similar to the U-Net, the Attention U-Net features an encoder-decoder architecture. The encoder captures hierarchical features through convolutional and pooling layers, while the decoder re-constructs the segmentation map using up-sampling and convolutional layers.

2. Skip Connections:

Skip connections facilitate the flow of information between corresponding encoder and decoder layers, aiding in the recovery of spatial details during the up-sampling process.

3. Attention Blocks:

Attention blocks are strategically inserted within the network. These blocks typically consist of self-attention mechanisms or channel-wise attention mechanisms, allowing the model to selectively emphasize or suppress specific regions or channels in the feature maps.

4. Self-Attention Mechanism:

The self-attention mechanism enables the model to weigh the importance of different positions in the feature maps, capturing long-range dependencies and improving the model's ability to recognize intricate patterns.

5. Channel-Wise Attention:

Channel-wise attention focuses on enhancing specific channels in the feature maps based on their relevance to the segmentation task. This helps the network adaptively weight different channels for improved contextual understanding.

6. Multi-Scale Context:

Attention U-Net excels in capturing multi-scale contextual information, allowing it to recognize both local and global features. This is particularly beneficial for image segmentation tasks where accurate delineation of object boundaries requires a nuanced understanding of context.

7. Final Layers:

The final layers of the Attention U-Net process the refined feature maps and generate the segmentation output. This output represents a highly context-aware and accurately weighted segmentation map.

In summary, the Attention U-Net augments the U-Net architecture by integrating attention mechanisms, offering an enhanced capacity to capture contextual information and weigh features adaptively. This architecture is particularly powerful in tasks demanding precise segmentation, such as medical image analysis and various computer vision applications.

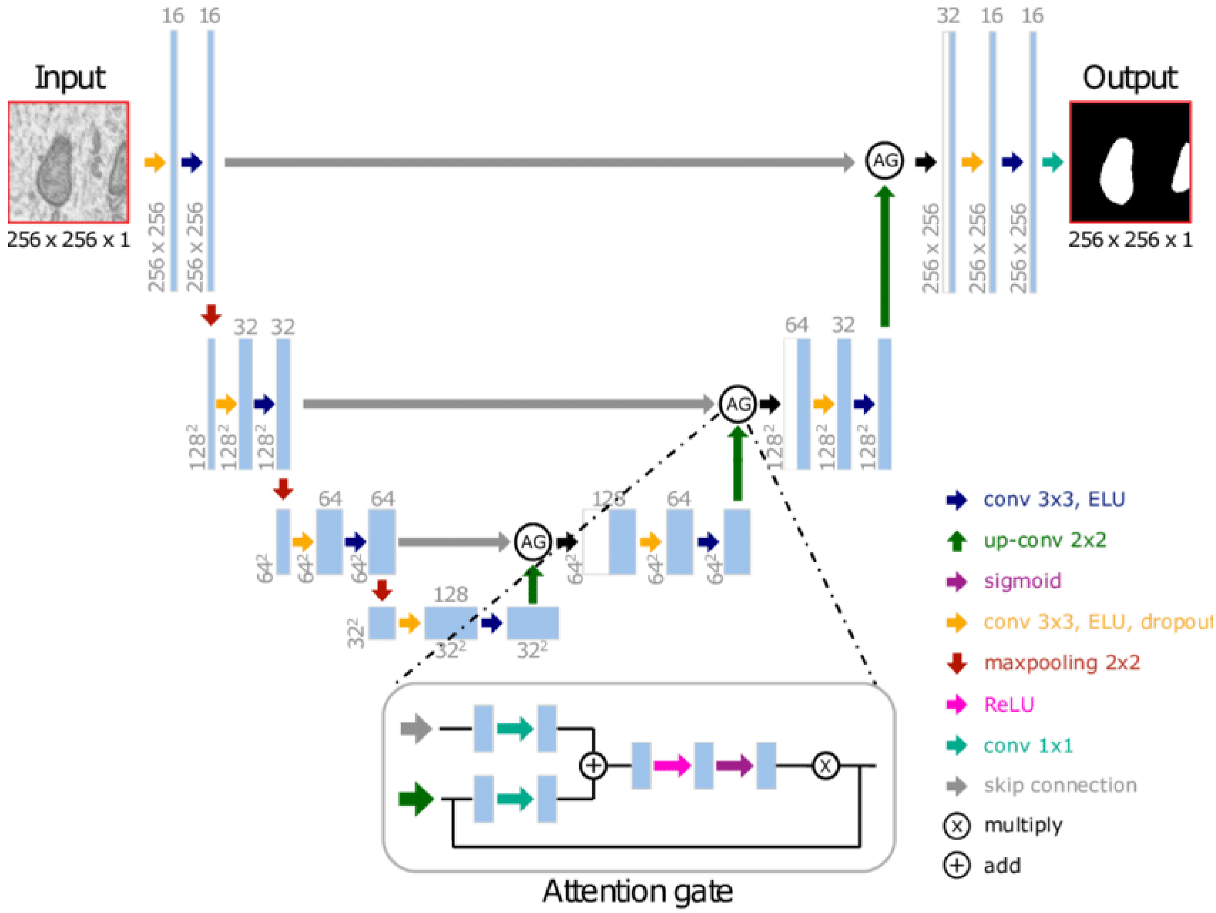


Fig: Attention UNet Architecture

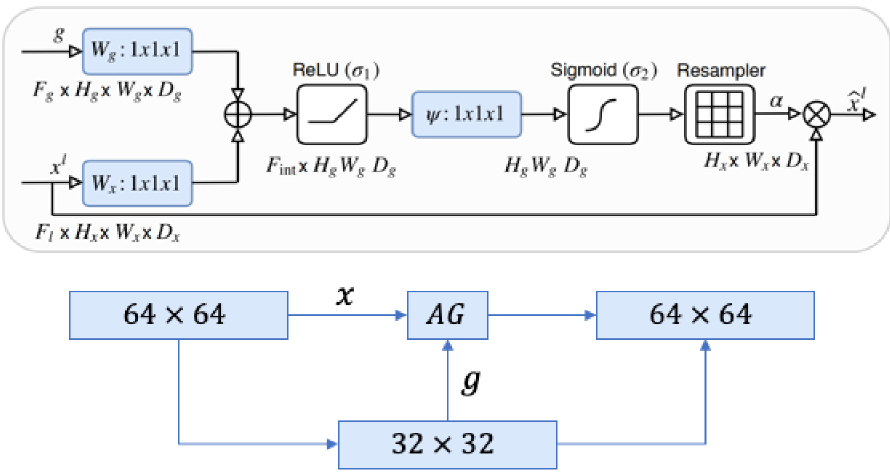


Fig: Attention Gate

Model Training, Validation and Hyperparameters:

When trained on **Slum Dataset** and checked for other hyperparameters.

- Number of_Epochs:15
- Batch size: 16
- Learning Rate: 1e-4
- Image Height: 256
- Image Width: 256
- Train Size: 2838
- Validation Size: 710

Metrics Table, Graphs and Masks:

Tables:

Training:

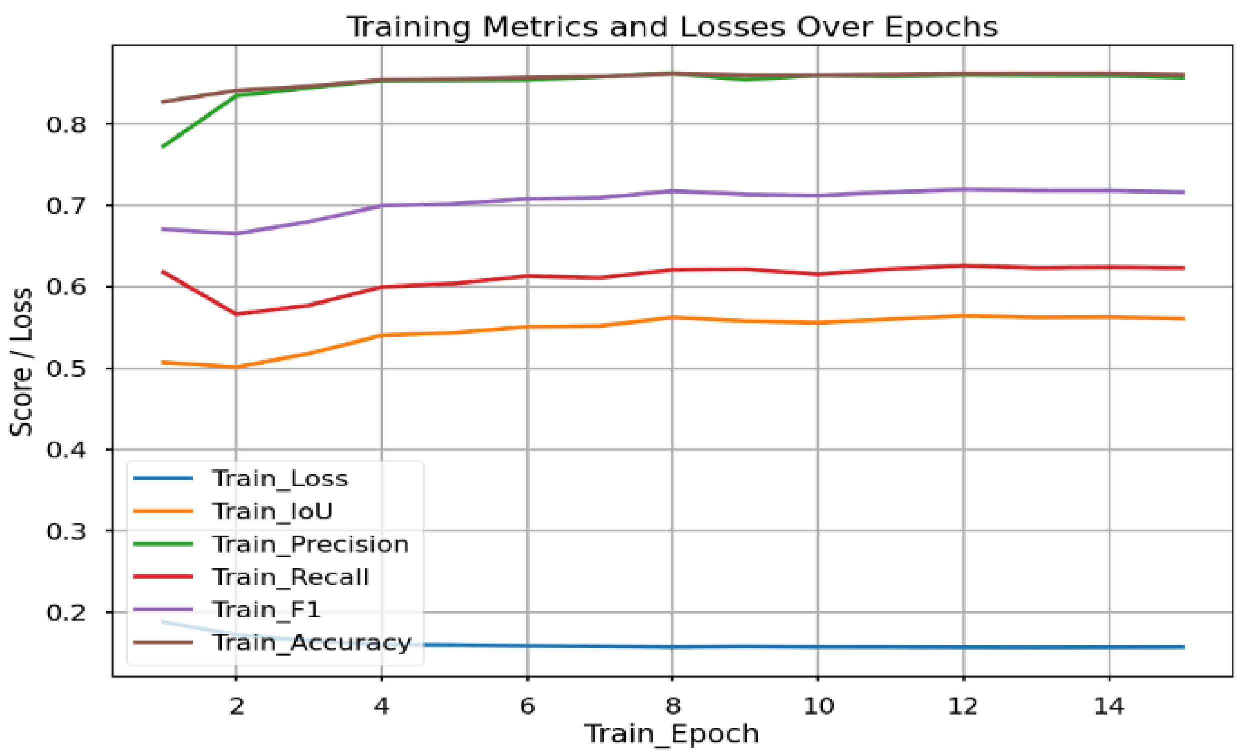
Training	Final Val	Average Val
Loss	0.15723162018850947	0.16131170212329551
IOU	0.5606082027732056	0.5464876051732577
Precision	0.8570355844926095	0.8494574502943111
Recall	0.6231074374931324	0.6109451317911052
F1-Score	0.716276563513972	0.7043718201594565
Accuracy	0.8603439761182639	0.855056635807567

Validation:

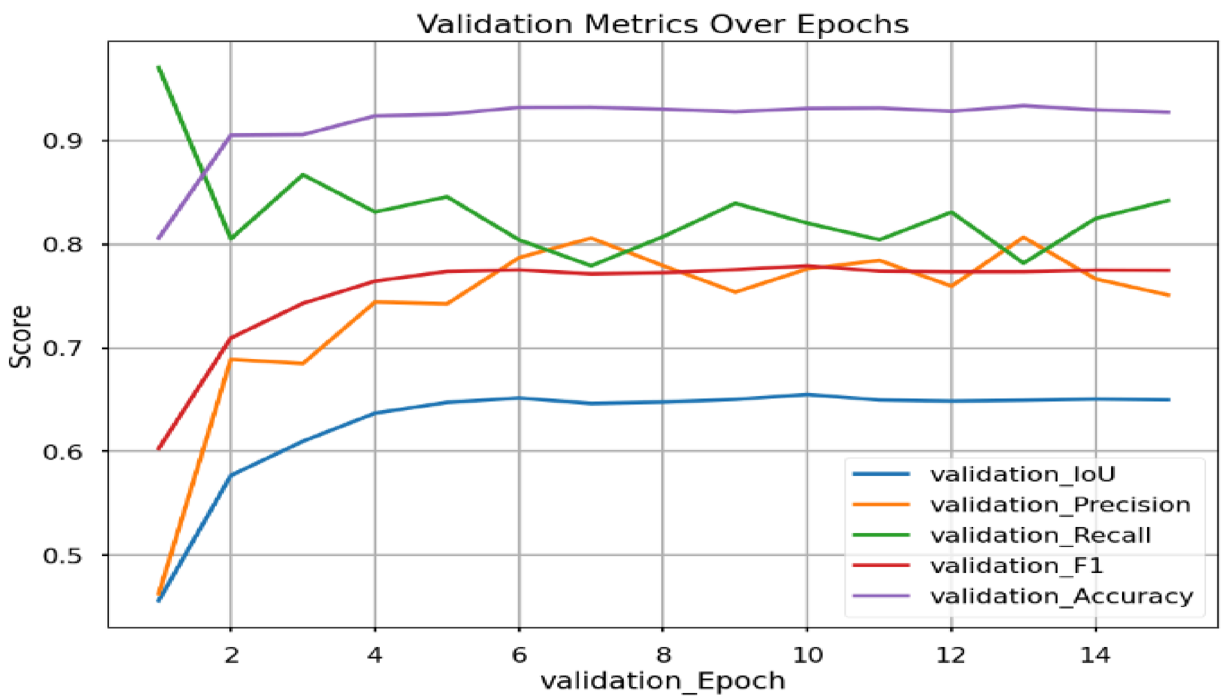
Validation	Final Val	Average Val
F1_Score	0.7746751661103345	0.7557594341583403
IOU	0.6498457509905271	0.628406179051552
Precision	0.7510247080409074	0.7395086373179571
Recall	0.8419589992671852	0.8301579786874985
Accuracy	0.9273315644600022	0.9178314166001871

Plots:

Training Metrics:



Validation Metrics:



Original vs Predicted Mask:

Original Mask:

Predicted Mask:

Chapter 6: Conclusion

In conclusion, this thesis introduced many deep learning model designed for the detection and segmentation of informal settlements in satellite imaginary, leveraging very high-resolution optical imagery. The model demonstrated noteworthy performance, achieving an **F1 score of 78% and accuracy of 93%**, surpassing the efficacy of prior state-of-the-art methods in the domain. These outcomes underscore the capability of deep learning models in accurately discerning informal settlements, thereby offering significant insights for urban planning and policy formulation. Out of all the models **ResNet50** as an encoder **pretrained on ImageNet dataset** achieved the best metrics among UNet from scratch, HRNET, Attention UNET with loss as Focal and Adam optimizer.

The dataset employed in this study was of high quality, meticulously labeled by the us to ensure precision in training the model. Opting for a U-Net architecture with pre-trained weights proved to be apt for the task, and careful fine-tuning of hyperparameters further optimized the model's performance.

The practical implications of this model are substantial, offering urban planners and policymaker's valuable insights into the distribution and attributes of informal settlements in Mumbai. These findings can inform decision-making processes related to infrastructure planning, housing policies, and the provision of essential services for these communities.

While the study's results are promising, opportunities for enhancement and future exploration persist. Scaling the model to cover larger areas and diverse urban environments represents a potential avenue for improvement. Additionally, integrating supplementary data sources, such as satellite imagery and socio-economic data, holds the potential to provide a more holistic understanding of the characteristics of informal settlements in Mumbai.

This thesis underscores the potential of deep learning models in addressing intricate urban challenges and contributes significantly to the realm of slum detection and segmentation. Several recommendations emerge from the study's outcomes and limitations, aimed at refining the model's accuracy and scalability.

Firstly, enhancing the dataset's size and diversity can fortify the model's robustness and mitigate overfitting. Integrating varied data types, such as satellite imagery and street-level photographs, offers the potential to furnish additional contextual information, augmenting the model's accuracy.

Secondly, exploring alternative deep learning architectures, including Fully Convolutional Networks (FCNs) and U-Net, HRNET, Attention UNet, holds promise for refining segmentation performance and reducing computation time. Fine-tuning hyperparameters like learning rates, filter quantities, and activation functions further stands out as a potential avenue for improvement.

Thirdly, scaling the model to encompass larger geographical areas, such as entire cities, stands as a crucial step toward providing invaluable insights for urban planning and policymaking. However, this necessitates the development of efficient algorithms and infrastructure to handle substantial data volumes.

Lastly, adapting the model to diverse urban contexts, such as identifying green spaces or estimating population density, holds significant potential for broader social and economic impacts. Implementing these recommendations collectively aims to foster a more accurate and scalable model, extending its applications in the expansive field of urban planning and development.

Chapter 6: Future Work

Future work will be more focused on the implementation of new model which results better accuracy, Precision, F1 score and IOU on our dataset. We will train our model on new dataset or incorporating additional data sources, such as socioeconomic data with high resolution. It would help to better understand the socio and economic factors that contribute to the development of informal settlements.

An additional promising direction for future research involves scaling the model to cover expansive areas, including entire cities, leveraging advanced computing resources and cloud based solutions. This approach has the potential to offer a more holistic understanding of informal settlements and their distribution across urban landscapes.

Furthermore, exploring the adaptability of the model to various urban contexts, such as rural areas and developing countries, presents an avenue for extended research. Given the potential differences in characteristics and spatial distributions of informal settlements in these settings, this exploration might require the creation of new training datasets and the modification of the model architecture to better align with the nuances of these environments.

References

[1] Cao, Kaili, and Xiaoli Zhang. "An improved res-unet model for tree species classification using airborne high-resolution images." *Remote Sensing* 12.7 (2020): 1128.

[2] Luo, Zhongming, et al. "Micro-vessel image segmentation based on the AD-UNet model." *IEEE Access* 7 (2019): 143402-143411.

[3] Wu, Zifeng, Chunhua Shen, and Anton Van Den Hengel. "Wider or deeper: Revisiting the resnet model for visual recognition." *Pattern Recognition* 90 (2019): 119-133.

[4] Khan, Riaz Ullah, et al. "Evaluating the performance of resnet model based on image recognition." *Proceedings of the 2018 International Conference on Computing and Artificial Intelligence*. 2018.

[5] Yu, Changqian, et al. "Lite-hrnet: A lightweight high-resolution network." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2021.

[6] Wu, Huisi, et al. "Optimized HRNet for image semantic segmentation." *Expert Systems with Applications* 174 (2021): 114532.

[7] Trebing, Kevin, Tomasz Stańczyk, and Siamak Mehrkanoon. "SmaAt-UNet: Precipitation nowcasting using a small attention-UNet architecture." *Pattern Recognition Letters* 145 (2021): 178-186.

[8] Thomas, Edwin, et al. "multi-res-attention UNet: a CNN model for the segmentation of focal cortical dysplasia lesions from magnetic resonance images." *IEEE Journal of Biomedical and Health Informatics* 25.5 (2020): 1724-1734.

[9] Bharati, Subrato, et al. "CO-ResNet: Optimized ResNet model for COVID-19 diagnosis from X-ray images." *International Journal of Hybrid Intelligent Systems* 17.1-2 (2021): 71-85.

[10] Cinar, Necip, Alper Ozcan, and Mehmet Kaya. "A hybrid DenseNet121-UNet model for brain tumor segmentation from MR Images." *Biomedical Signal Processing and Control* 76 (2022): 103647.

[11] X. Wang, S. Liu, P. Du, H. Liang, J. Xia, and Y. Li, “Object-based change detection in urban areas from high spatial resolution images based on multiple features and ensemble learning,” Remote Sensing, vol. 10, no. 2, 2018.

[12] Trebing, Kevin, Tomasz Stańczyk, and Siamak Mehrkanoon. "SmaAt-UNet: Precipitation nowcasting using a small attention-UNet architecture." *Pattern Recognition Letters* 145 (2021): 178-186.

[13] Li, Ying, et al. "A-hrnet: Attention based high resolution network for human pose estimation." *2020 Second International Conference on Transdisciplinary AI (TransAI)*. IEEE, 2020.

[14] Anderson, P., He, X., Buehler, C., Teney, D., Johnson, M., Gould, S., Zhang, L.: Bottom-up and top-down attention for image captioning and vqa. arXiv preprint arXiv:1707.07998 (2017)